Chi Lung (Kevin) Ma

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EDUCATION

University of Washington (UW) — Seattle, WA
M.S. in Mechanical Engineering (Data Science Option), GPA: 3.83/4.0 — Sep. 2025 - Expected Jun. 2027

National Yang Ming Chiao Tung University (NYCU) — Taiwan
B.S. in Mechanical Engineering — Sep. 2018 - Jun. 2022

EXPERIENCE

Research Assistant — Seattle, WA
Robot Learning Lab, UW Paul G. Allen School of CSE — Sep. 2026 - Present

- Investigating diffusion-based approaches to address covariate shift in reinforcement learning for robotic manipulation.

Robot Learning Research Intern | Robotics Systems Team Lead — Seattle, WA
Corvinus Labs — Jun. 2026 - Aug. 2026

- Architected and implemented an end-to-end robot learning research platform, connecting Isaac Sim, LeRobot data pipelines, Policy training, and closed-loop evaluation to support reproducible policy experiments.
- Evaluated four data-scaling configurations over 200 episodes each, revealing that offline action metrics failed to predict closed-loop task success and motivating a shift toward task-level evaluation.
- Traced a 43% closed-loop abort rate through controlled ablations to infeasible commanded rotations at the IK layer, informing an SE(3) backtracking fix that lifted task success from 58% to 73%.
- Built a unified simulation-to-hardware control interface for a 6-DoF robotic arm, supported by hardware-free tests; improved trajectory replay accuracy from 10.0 mm to 3.4 mm through servo tuning.

Autonomy Student Researcher — Seattle, WA
Washington Aerial Robotics, UW — Jan. 2026 - Apr. 2026

- Designed and prototyped a mission-level strategy for multi-drone exploration using frontier-based planning and confidence-weighted mapping, validating basic feasibility under partial observability.

Perception Team Lead — Taiwan
ORCA Autonomous Underwater Vehicle, NYCU — Jun. 2020 - Sep. 2022

- Led a perception team in developing and deploying a ROS-based YOLO detection pipeline on an autonomous underwater vehicle, supporting underwater object recognition for a team that advanced past the SAUVC preliminary round.

PROJECTS

Spatial Generalization of VLA Manipulation under Distribution Shift

- Evaluated spatial generalization in VLA manipulation by systematically perturbing initial end-effector poses, revealing a sharp drop in SmolVLA task success from 90% at a 5 cm offset to 20% at 10 cm.
- Compared against a hierarchical vision-language policy with explicit 3D grounding, which retained 80% success at the same 10 cm shift, suggesting improved robustness through geometric reasoning.

Sim2Act — Robot Learning Evaluation Pipeline in NVIDIA Isaac Lab

- Built a scalable simulation-to-policy pipeline using a PPO teacher across 4,096 parallel Isaac Lab environments to generate multimodal demonstrations and evaluate ACT policies under closed-loop distribution shifts.
- Identified a visual shortcut in a learned pushing policy through controlled camera ablations, distinguishing data-induced failure from insufficient model capacity.

Quantization Robustness of Multimodal LLMs on Edge GPUs

- Benchmarked seven TensorRT-LLM quantization configurations for Qwen2-VL-2B across VQAv2, POPE, and MME, evaluating accuracy, latency, and memory trade-offs for edge deployment.
- Isolated FP8 KV-cache quantization as the source of an 8.8-point VQAv2 accuracy regression through controlled ablations, revealing disproportionate degradation in fine-grained visual matching.

SKILLS

Programming & Tools: Python, C++, Git, Linux
Machine Learning: Diffusion Policy, Imitation Learning, Reinforcement Learning, LLM, VLM, VLA Models
Frameworks: PyTorch, LeRobot, NVIDIA Isaac Sim/Lab, MuJoCo, ROS, TensorRT-LLM